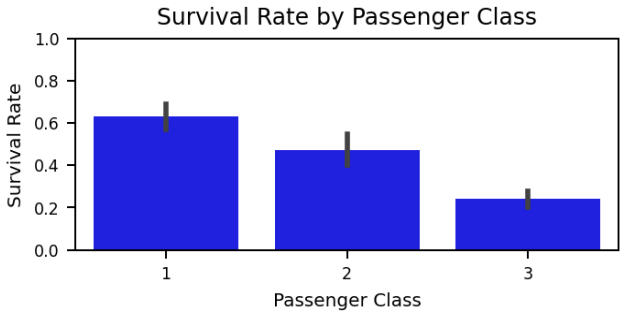
Assignment 2 – Titanic Data Analysis Final Report

Dataset: Titanic training data with 891 passenger records. The overall survival rate was 38.38%. Survived = 0 means did not survive; Survived = 1 means survived.

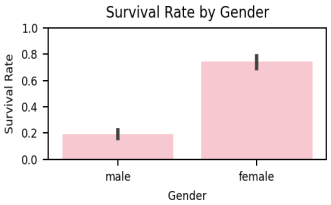
Hypothesis One: Survival Rate vs. Passenger Class

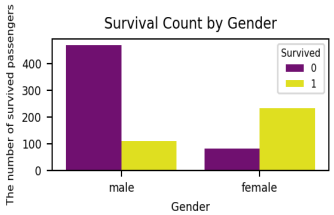
The survival rate was associated with passenger class. First-class passengers had the highest survival rate at 63% (mean = 0.63), followed by second-class passengers at 47% (mean = 0.47), while third-class passengers had the lowest rate at 24% (mean = 0.24). The blue bar chart shows that survival rates decreased from first class to third class.



Hypothesis Two: Survival Rate vs. Passenger Gender

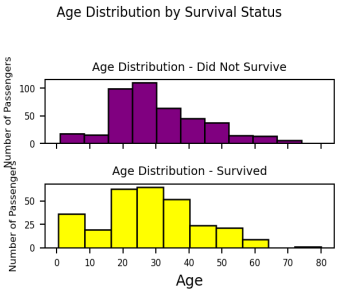
Female passengers had a survival rate of 74% (mean = 0.74), compared with 19% (mean = 0.19) for male passengers. The pink bar chart compares survival rates, while the purple-and-yellow count plot shows the number of passengers in each survival group. In this dataset, more females survived, while more males did not survive.

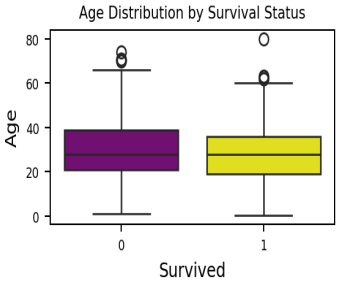




Hypothesis Three: Survival Rate vs. Age

The average age was 30.6 years for non-survivors and 28.3 years for survivors. The purple-and-yellow histograms show the age frequencies for the two groups. The boxplot compares their medians and spread; the median ages were 28.0 and 28.0 years, respectively. The age distributions overlap considerably, suggesting that the difference in age between survivors and non-survivors is relatively small overall.





Overall Conclusion

Passenger class and gender showed clear differences in survival rates. First-class passengers and female passengers had higher observed survival rates. For age, the average ages and median ages of survivors and non-survivors were relatively close, and the two distributions overlapped. Therefore, age showed a less clear relationship with survival than passenger class and gender. These results describe associations in this dataset and do not prove causation.